

Exercise 3.7

$$a) \frac{8.769 \times \sqrt[3]{27.03}}{24.51}$$

$$\lg 8.769 + \frac{1}{3} \lg 27.03 - \lg 24.51$$

$$= 0.9428 + \frac{1}{3} [1.4320] - 1.3893$$

$$= 0.9428 + 0.4773 - 1.3893$$

$$= 0.0308 \times 10^2$$

$$x = \text{Anti} \lg 0.0308$$

$$= 1.073$$

$$b) \frac{\sqrt{9.18 \times 8.02^2}}{4.83}$$

$$\lg = \frac{1}{2} 0.9628 + \frac{2}{2} 0.9041 - 0.4426$$

$$\lg \frac{1}{2} 9.18 + 8.02^2 - \lg 4.83$$

$$= \frac{1}{2} 0.9628 + 1.8082 - 0.4426$$

$$= \frac{1}{2} 2.7710 - 0.4426$$

$$= 1.3855 - 0.4426$$

$$= 0.3929 \times 10^2$$

$$\text{Anti} \lg 0.3929$$

$$= 2.47$$

$$d. \frac{3 \times 0.752^2}{\sqrt{17.46}}$$

$$\lg 3 + 2 \lg 0.752 - \frac{1}{2} \lg 17.46$$

$$0.4771 + \overset{0}{\cancel{0.4771}} \cdot 1238 - \frac{1}{2} \times \cancel{1.2544}$$

$$0.4771 - 0.2476 - 0.6272$$

$$= -0.3977 \times 10^1$$

$$\text{Anti } \lg = -0.3977$$

$$= 0.400$$

$$F. \frac{3.291 \times \sqrt[3]{0.0234}}{0.8415}$$

$$\lg 3.291 + \frac{1}{3} \lg 0.0234 - \lg 0.8415$$

$$= 0.5120 + \frac{1}{3} \lg 2.34 - 1.6308 - 0.0444$$

$$= \cancel{0.5120} - \cancel{0.5436} + \cancel{0.0444}$$

$$= \cancel{0.0183} = 0.5120 - 0.5436 + 0.0444$$

$$= 0.0183$$

$$\text{Anti } \lg = 0.0183 \times 10^2 = 1.83$$

$$= 1.83$$